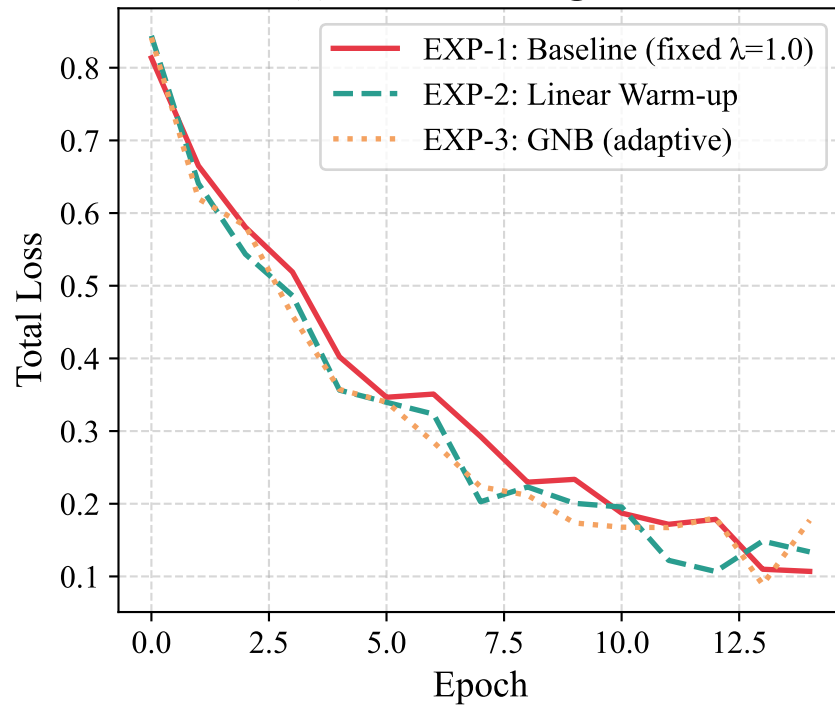


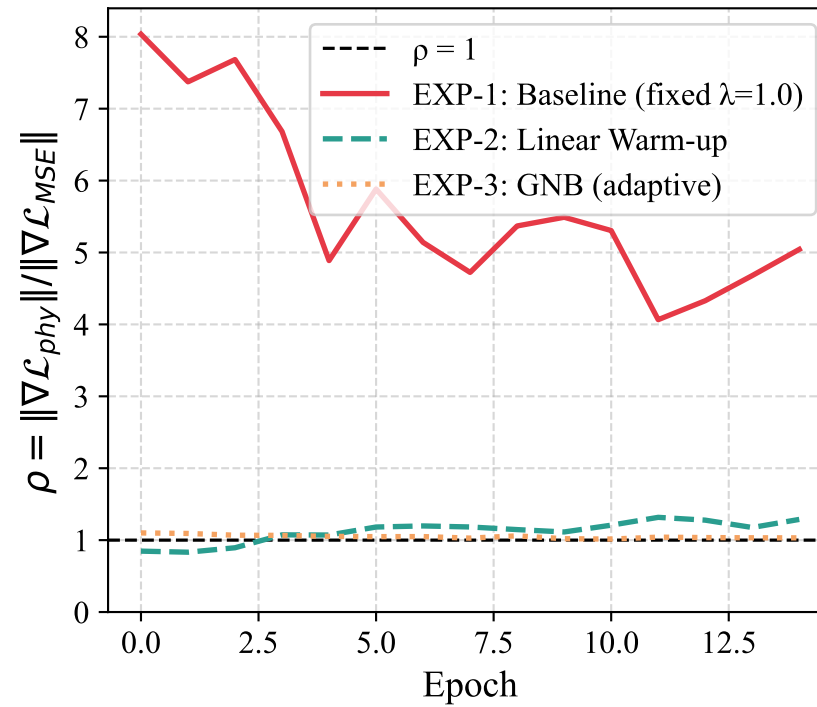
Adaptive Physics-Constrained Loss Scheduling in HydroGraphNet

EXP-1: Fixed $\lambda=1.0$ | EXP-2: Linear Warm-up | EXP-3: GNB

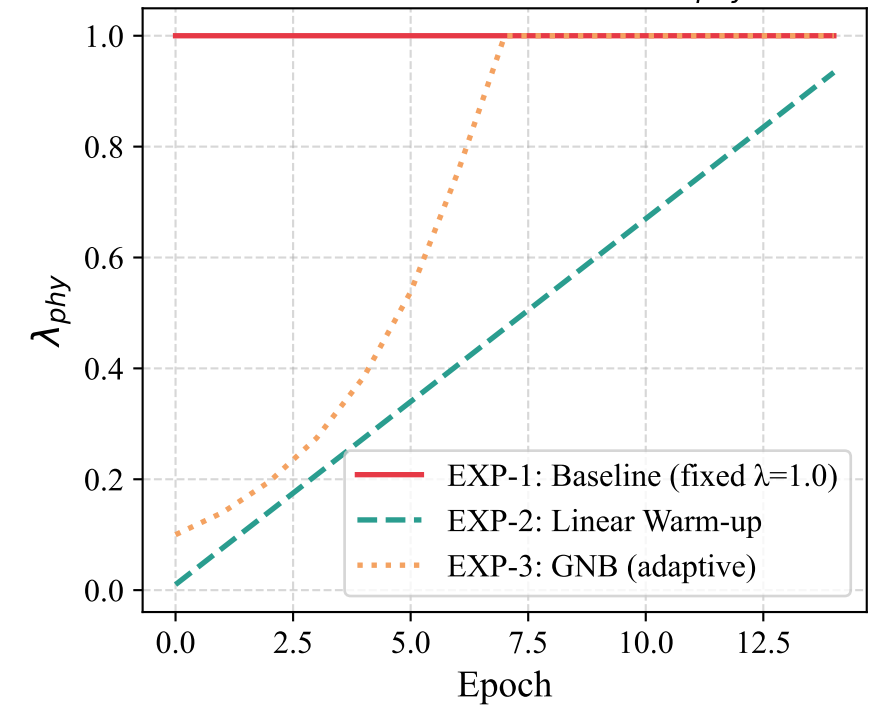
(a) Total Training Loss



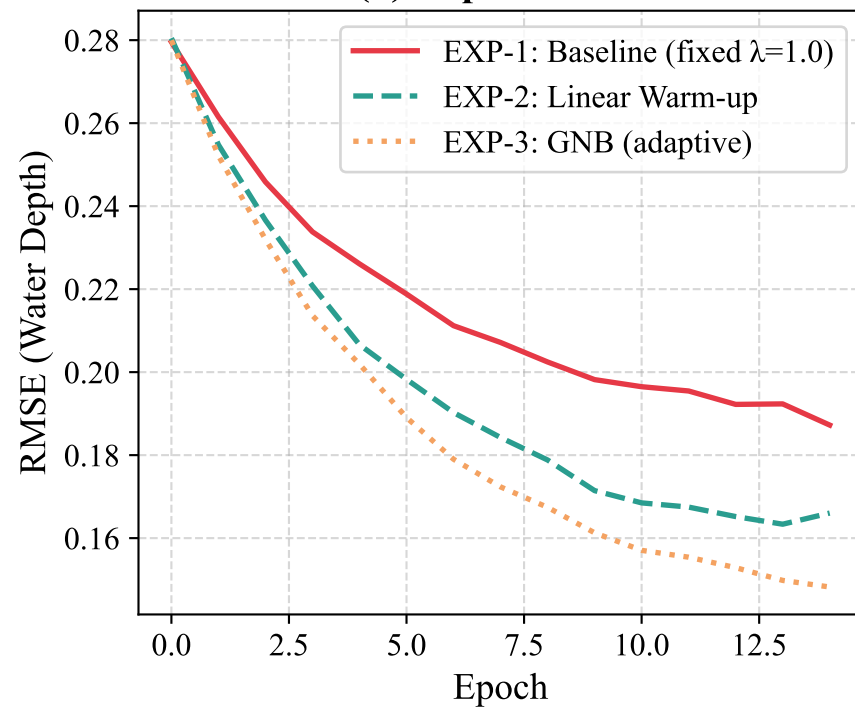
(b) Gradient Norm Ratio ρ



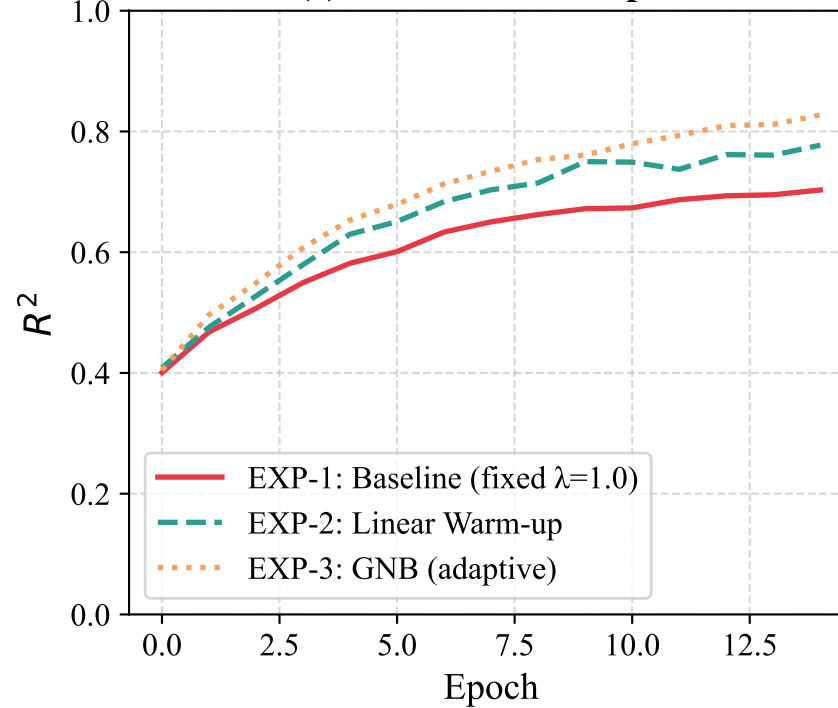
(c) Physics Loss Weight λ_{phy}



(d) Depth RMSE



(e) R^2 — Water Depth



(f) Final Epoch Comparison

